FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE

JEE (Main), 2015

PART TEST –IV

Time Allotted: 3 Hours **Maximum Marks: 432**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 03 and 10 to 12)** contains 6 multiple choice questions which have only one correct answer. Each question carries **+8 marks** for correct answer and **– 2 mark** for wrong answer.

Section-A (04 to 09 and 13 to 30) contains 24 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

Straight Objective Type

This section contains **30 multiple choice questions** numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

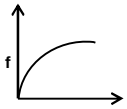
1. In a standard meter bridge experiment conducted to measure the specific resistance of a wire the following data was recorded:-
Length (L) $= (1 \pm .01)\text{m}$ measured by a meter scale
Radius of wire = 1mm $\pm$.01mm. (measured by screw gauge).
Resistance of wire (R) = (5$\pm$.01) Ω
The maximum possible error in the measurement of specific resistance is
(You may use the formula:- $\rho = \frac{\pi r^2}{L} R$)
(A) .1$\mu\Omega$m (B) .2$\mu\Omega$m
(C) .5$\mu\Omega$m (D) .6$\mu\Omega$m

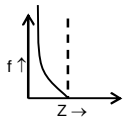
2. A physical quantity P is related to four measurable quantities as $P = \frac{a^3 b^2}{c^3 d^{4.}}$. If the errors in measurement of a, b, c, d are 0.1%, 0.2%, 0.3%, 0.1% respectively, what is the maximum percentage error in measurement of P.
(A) 1% (B) 2%
(C) 3% (D) 4%

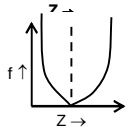
3. When N_0 molecules of a radioactive nuclide are taken at t = 0, the activity reduces to 'A' in time t_0. When $3N_0$ molecules of the same nuclide are taken, activity will become 3Ae in time t equal to
(A) t_0 (B) $t_0 + \frac{1}{\lambda}$
(C) $t_0 - \frac{1}{\lambda}$ (D) None of these
Where; λ is decay constant of the Nuclide

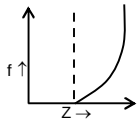
Space for rough work

4. The plot of characteristic frequency 'f' in case of K_y –X-rays, with the atomic number 'Z' of an element may be best represented by

(A) 

(B) 

(C) 

(D) 

5. Two waves of Intensity I_1 and I_2 interfere. The maximum intensity produced to the minimum intensity is in the ratio α. Then, I_1 / I_2 is equal to

(A) $\frac{\alpha-1}{\alpha+1}$, if $I_1 > I_2$

(B) $\frac{1+\alpha}{\alpha-1}$, if $I_1 > I_2$

(C) $\left(\frac{1+\sqrt{\alpha}}{\sqrt{\alpha}-1}\right)^2$, if $I_1 < I_2$

(D) $\left(\frac{\sqrt{\alpha}-1}{1+\sqrt{\alpha}}\right)^2$ if $I_1 > I_2$

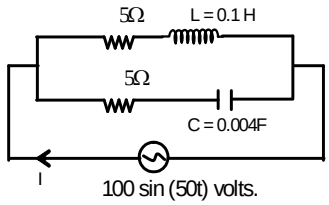
6. The current i in the circuit at any time t is

(A) 20 A

(B) $20\sin\left[50t+\frac{\pi}{4}\right]$ A

(C) $20\sin\left[50t\ \frac{-\pi}{4}\right]$ A

(D) $20\sin(50t)$ A



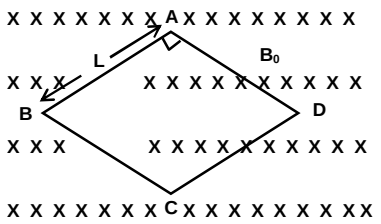
7. A conducting square loop of length 'L' is placed in a magnetic field of Induction B_0. End 'A' and End 'C' are moved away from each other. So that finally the square loop becomes a straight line. The total charge flowing through the loop, if its resistance is 'R' then

(A) $\frac{B_0L^2}{\pi R}$

(B) $< \frac{B_0L^2}{R}$

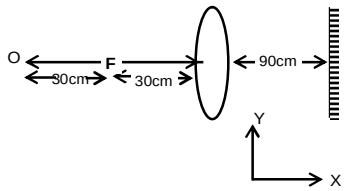
(C) $\frac{B_0L^2}{R}$

(D) $> \frac{B_0L^2}{R}$



Space for rough work

8. A thin biconvex lens of focal length 30cm is Kept 90 cm away from a plane mirror kept parallel to the plane of the lens. A point object is placed on the optical axis of the lens 60cm away from the lens as shown in the figure. Now, if the point object is slowly moved towards the lens with a small velocity, the image of the object in the mirror.

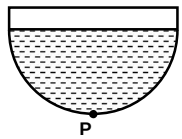


(A) Moves in the + ve x direction
(B) Moves in – ve x direction
(C) Does not move
(D) Data insufficient

9. The ratio of de-Broglie wavelength associated with an electron moving in nth orbit of a hydrogen atom and its first Bohr's radius is

(A) $2\pi n : 1$
(B) $\sqrt{n} : 2\pi$
(C) $n^3 : 1$
(D) $n : 1$

10. A hemispherical bowl of radius 10 cm is filled with liquid of refractive index $\mu = 4/3$. A glass plate of refractive index 1.5 is placed on the top of bowl. If for the observer above the plate the shift in position of a point P on the bottom is 3 cm find the thickness of glass plate.



(A) 1.5 cm
(B) 1 cm
(C) 7 cm
(D) 10 cm

11. An electron is knocked out from K shell of tungsten atom. P_1, P_2, P_3 are probabilities that an electron jumps from L, M and N shell to K shell, then which is correct

(A) $P_1 = P_2 = P_3$
(B) $P_3 > P_2 > P_1$
(C) $P_1 = P_2 > P_3$
(D) $P_1 > P_2 > P_3$

12. A potential difference of 2 kV is applied across a X-ray tube and X-rays thus produced fall on a metallic plate at a potential of 100 V. What is the maximum K.E. of photoelectrons emitted from the plate if its work function is 5.8 eV ?

(A) 2094.2 eV
(B) 2105.8 eV
(C) 1906.8 eV
(D) 1894.2 eV

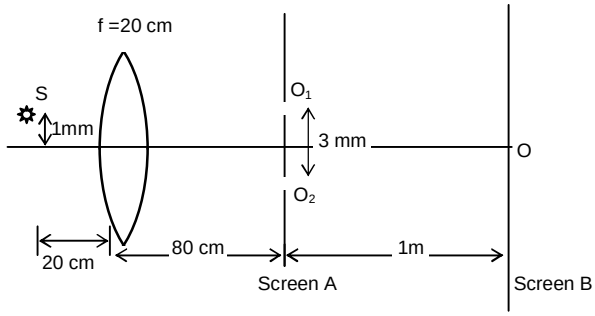
13. The velocity of a real object, moving along the principal axis of a lens, at an instant is 4 times the velocity of the virtual image for a lens. Then the ratio of focal length to image distance is

(A) 2 : 1
(B) 1 : 2
(C) 4 : 1
(D) 1 : 4

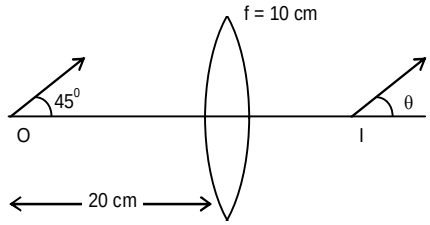
Space for rough work

14. There are 10 atoms of C^{14} in a sample of carbon atoms. After one half life period the number of radioactive atoms of C^{14} in sample will be
(A) exactly 5
(B) certainly more than 5
(C) certainly less than 5
(D) can not find exactly

15. An arrangement for YDSE is shown in figure. S is a light source then find the position of central maxima from point O on screen B.
(A) 5 mm below O
(B) 5 mm above O
(C) 30 mm below O
(D) 12 mm above O

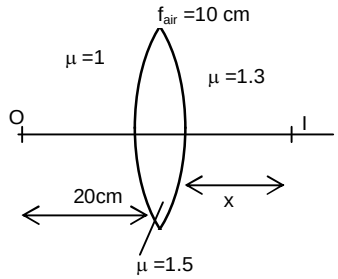


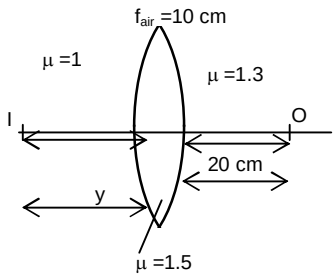
16. An object starts moving at an angle of 45^0 with the principal axis as shown in the figure in front of a biconvex lens of focal length + 10 cm. If θ denotes the angle at which image starts moving with principal axis then



(A) $\theta = \frac{3\pi}{4}$
(B) $\theta = \frac{\pi}{2}$
(C) $\theta = \frac{\pi}{4}$
(D) $\theta = -\frac{\pi}{4}$

17. A thin equiconvex lens, made of material of refractive index 1.5, has a focal length of 10 cm in air. One side of the lens is replaced by a medium (B) of refractive index 1.3. When a point object is placed in air at a distance of 20 cm from the lens on its principal axis, its image is formed at distance of x cm in the medium (B), from the lens. When the same object is placed in the medium B, at a distance of 20 cm from the lens on its principal axis, its image is formed at a distance of y cm in air, from the lens. Then

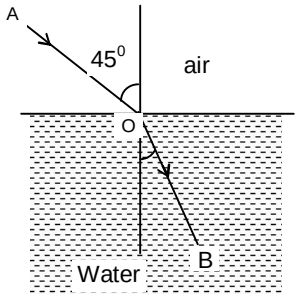




(A) x > 1.3 y
(B) x < 1.3 y
(C) x = 1.3 y
(D) none of the above is true.

Space for rough work

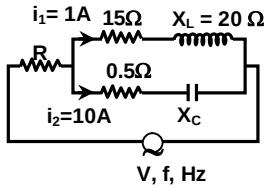
18. Planar light waves and sound waves incident on the air – water interface along the direction of the line are AO as shown in the figure. Assume that AO represents the direction of the incident ray (for both light and sound). Given that the speed of sound wave in air and water is 330 m/s and 1500 m/s respectively and of the light wave it is 3×10^8 m/s and 2.26×10^8 m/s. respectively then, (assume Snell's law is valid for both waves):



(A) light wave and sound wave will refract through same angle.
(B) angle of refraction for light wave will be 30^0.
(C) water is acoustically denser medium than air for sound wave.
(D) the angle between the two rays when they get separated by the air water surface is

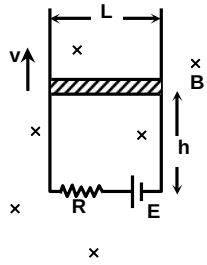
$$\left\{\frac{3\pi}{4} - \sin^{-1}\left(\frac{0.75}{\sqrt{2}}\right)\right\} \text{rad}..$$

19. Find the value of reactance X_C as shown in the figure.



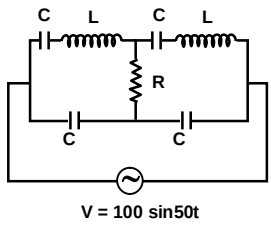
(A) 3Ω
(B) $\sqrt{6}\ \Omega$
(C) 2Ω
(D) none of these

20. A conducting cross bar of mass m and length L can slide, without friction, vertically up and down on two conducting rails. The rails are connected on the bottom via a variable resistance R and a battery E, so that rails and the bar form a rectangular conducting loop. A uniform magnetic field B points into the plane of the loop. Whole system is placed in constant gravitational field g. Find the value of E for which rod will move up with a constant velocity v.



(A) $\frac{mgR}{\ell B}$
(B) $\frac{mg}{R\ell B} - B\ell V$
(C) $\frac{mg}{R\ell B} + B\ell V$
(D) zero

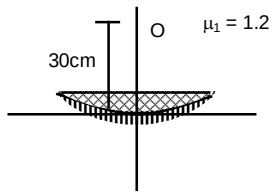
21. The value of the current in the resistor R in the following circuit will be (Given C = 1 μF, L = 1 H, R = 1 Ω)



(A) zero
(B) $\frac{100}{R}\sin(50t)$
(C) 100/R
(D) can't be calculated

Space for rough work

22. A concave mirror is placed in a medium of refractive index $\mu_1 = 1.2$. An object placed at 30 cm from pole of the mirror and its image coincides. Now a liquid of refractive index μ is placed in the concave mirror in the shaded region while the medium of refractive index 1.2 is retained in the remaining region. It is found that object needs to be shifted by 7.5 cm below O for the object and image to coincide. The refractive index of the liquid μ is



(A) $\frac{4}{3}$ (B) 1.2

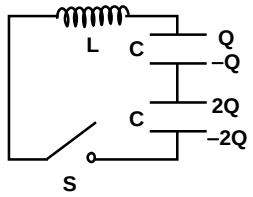
(C) 1.6 (D) none of the above

23. An object is placed at 30 cm from a convex lens of focal length 15cm. On the other side of the lens a convex mirror of focal length 12cm is placed so that the principal axis of both coincide. It is observed that the object and image coincide. What is the separation between the lens and mirror?

(A) 6 cm (B) 30 cm

(C) either (A) or (B) (D) none of the above

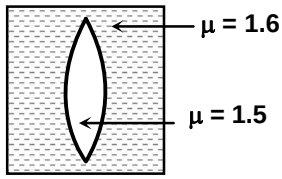
24. Initially two capacitors each of capacitors C have charges as shown and the inductor L has zero current. After the switch S is closed, the maximum current in the inductor is



(A) $\frac{3Q}{\sqrt{2LC}}$ (B) $\frac{3Q}{\sqrt{LC}}$

(C) $\frac{4Q}{\sqrt{LC}}$ (D) $\frac{5Q}{\sqrt{LC}}$

25. The given figure shows a convergent lens placed inside a cell filled with liquid. The lens has focal length +20 cm when in air, and its material has refractive index 1.50. If the liquid has refractive index 1.60, then focal length of the lens in the cell is

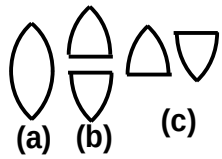


(A) 80 cm (B) –80 cm

(C) –24 cm (D) –160 cm

Space for rough work

26. An equiconvex glass lens (a) has focal length f and power P. It is cut into two symmetrical halves (b) by a plane containing the principal axis. The two pieces are recombined as shown in figure (c). The power of the new combination is



(A) zero (B) $\frac{P}{2}$

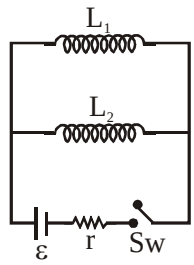
(C) 2P (D) none of the above

27. Choose incorrect statements:
An inductor of inductance L, capacitance C and resistance R are connected in series to a variable sinusoidal voltage supply, whose frequency is variable, but rms voltage is constant. At resonance
(I) Average power will be maximum (II) Average power will be minimum
(III) Impedence will be maximum (IV) current and voltage will be in same phase

(A) I and IV (B) II and III
(C) I and III (D) II, III, and IV

28. Which of the following is accompanied by the characteristic X-ray emission?
(A) α-particle emission (B) Electron emission
(C) Positron emission (D) K-electron capture

29. The switch Sw of the circuit shown in the figure is closed at t = 0. i_1 and i_2 are the electric current in the branch containing two ideal inductors L_1 and L_2 respectively, at a certain instant. Choose the most appropriate statement.



(A) At t = 0,(just after the switch is closed) $\frac{di_1}{dt} = \frac{\varepsilon}{L_1}$ and $\frac{di_2}{dt} = \frac{\varepsilon}{L_2}$

(B) At t = 0, ,(just after the switch is closed) $i_1 = 0, i_2 = 0$

(C) After long time $i_1 = \frac{\varepsilon L_2}{(L_1 + L_2)}$ and $i_2 = \frac{\varepsilon L_1}{(L_1 + L_2)r}$

(D) All the above are true

30. Monochromatic light is used in Young's double slit experiment. When one of the slits is covered by a transparent sheet of thickness 1.8×10^{-5} m, made of material of refractive index μ_1 number of fringes which shift is 18. when another sheet of thickness 3.6×10^{-5} m, made of material of refractive index μ_2 is used, number of fringes which shift is 9. Relation between μ_1 and μ_2 is given by

(A) $4\mu_2 - \mu_1 = 3$ (B) $4\mu_1 - \mu_2 = 3$

(C) $3\mu_2 - \mu_1 = 4$ (D) $2\mu_1 - \mu_2 = 4$

Space for rough work

Chemistry PART – II

SECTION – A

Straight Objective Type

This section contains **30 multiple choice questions** numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. If a metal crystal exists in fcc and bcc having the radius r. What is the edge length ratio (a_1/a_2) in two crystal structure, a_2, a_1 are the edge lengths of the cube in fcc and bcc respectively
   (A) $\sqrt{3}:\sqrt{2}$ (B) $\sqrt{2}:\sqrt{3}$
   (C) $\sqrt{6}:\sqrt{1}$ (D) $1:\sqrt{6}$

2. What is the oxidation number and coordination number of metal in $K_3\left[Fe\left(C_2O_4\right)_3\right]$
   (A) O.N = +2, C.N = 3 (B) O.N = +3, C.N = 3
   (C) O.N = +3, C.N = 6 (D) none of these

3. Orange red vapours of X is formed by heating the chloride salt with $K_2Cr_2O_7$ and concentrated H_2SO_4. When X on passing in NaOH gives yellow solution of Na_2CrO_4. Then X =
   (A) CrO_3 (B) $CrCl_3$
   (C) CrO_2Cl_2 (D) H_2CrO_4

4. The emf of the following three galvanic cells:
   I. $Zn/Zn^{2+}(1\,m)||Cu^{2+}(1\,m)|Cu$
   II. $Zn/Zn^{2+}(0.1\,m)||Cu^{2+}(1\,m)|Cu$
   III. $Zn/Zn^{2+}(1\,m)||Cu^{2+}(0.1\,m)|Cu$
   Are represented by E_1, E_2 and E_3. Which of the following statements is true?
   (A) $E_1 > E_2 > E_3$ (B) $E_3 > E_2 > E_1$
   (C) $E_3 > E_1 > E_2$ (D) $E_2 > E_1 > E_3$

5. In which of the following reactions, there is no change in valency?
   (A) $4KClO_3 \longrightarrow 3KClO_4 + KCl$ (B) $SO_2 + 2H_2S \longrightarrow 2H_2O + 3S$
   (C) $BaO_2 + H_2SO_4 \longrightarrow BaSO_4 + H_2O_2$ (D) $2BaO + O_2 \longrightarrow 2BaO_2$

6. The case of liquefaction of noble gases decreases in the order
   (A) $He > Ne > Ar > Kr > Xe$ (B) $Xe > Kr > Ar > Ne > He$
   (C) $Kr > Xe > He > Ar = Ne$ (D) $Ar > Kr > Xe > He > Ne$

Space for rough work

7. Which of the following is used as a foaming agent in froth floatation process is
(A) pine oil (B) amyl xanthenes
(C) $CuSO_4$ (D) KCN

8. The plot of $1/x_A$ versus $1/y_A$ (where x_A and y_A are the mole fractions of A in liquid and vapour phases respectively) is linear whose slope and intercept respectively are given as
(A) $P_A^0/P_B^0, \left(P_A^0 - P_B^0\right)/P_B^0$ (B) $P_A^0/P_B^0, \left(P_B^0 - P_A^0\right)/P_B^0$
(C) $P_B^0/P_A^0, \left(P_A^0 - P_B^0\right)/P_B^0$ (D) $P_B^0/P_A^0, \left(P_B^0 - P_A^0\right)/P_B^0$

9. K_{sp} of $AgBr = 8 \times 10^{-13}$
$Ag, AgNO_3\left(1.0M\right)//KBr\left(1.0M\right)AgBr, Ag$
The emf of the cell is
(A) -0.71 (B) +0.42
(C) -0.42 (D) +0.71

10. The voltage of a certain cell at 25°C and 20°C are 0.3525 and 0.3533 V respectively. If the number of electrons involved in the overall reaction are two, what is the value of ΔS^0 at 25°C
(A) 30.88 JK^{-1} (B) 60.11 JK^{-1}
(C) -30.88 JK^{-1} (D) -60.11 JK^{-1}

11. Which of the following is/are paramagnetic?
$NO_2, NO, N_2O_4, N_2O_2, N_2O_5$
(A) only NO_2 (B) NO_2, NO
(C) NO, NO_2, N_2O_5 (D) all are paramagnetic

12. $KClO_4$ is made in the following sequence of reactions:
$Cl_2 + KOH \longrightarrow KCl + KClO + H_2O$ ...(i)
$KClO \longrightarrow KCl + KClO_3$ ...(ii)
$KClO_3 \longrightarrow KClO_4 + KCl$ ...(iii)
What mass of Cl_2 is needed to produce 1.0 kg of $KClO_3$?
(A) 2.05 kg (B) 1 kg
(C) 4.02 kg (D) 5 kg

13. A solution containing 6 g urea per litre isotonic with a solution containing
(A) 5.85 g NaCl per litre (B) 34.2 g sucrose per lire
(C) 36.0 g glucose per litre (D) 6.0 g of sucrose per litre

Space for rough work

14. In the sequence of reaction

$A \xrightarrow{k_1} B \xrightarrow{k_2} C \xrightarrow{k_3} D$

$k_3 > k_2 > k_1$, then the rate determining step of the reaction is

(A) $A \rightarrow B$ (B) $B \rightarrow C$

(C) $C \rightarrow D$ (D) $A \rightarrow D$

15. Chemically rust of iron is

(A) hydrated ferrous oxide

(B) hydrated ferric oxide

(C) only ferric oxide

(D) a mixture of ferric oxide with a little of $Fe(OH)_3$

16. Which of the following will give four isomers?

(A) $\left[Co(en)(NH_3)_2Cl_2\right]Cl$ (B) $\left[Co(PPh_3)(NH_3)_2Cl_2\right]Cl$

(C) $\left[Co(en)_3\right]Cl_3$ (D) $\left[Co(en)_2Cl_2\right]Br$

17. The colourless species is

(A) VCl_3 (B) $VOSO_4$

(C) Na_3VO_4 (D) $\left[V(H_2O)_6SO_4\right]H_2O$

18. At 25°C, the standard emf of a cell having reaction involving two electron exchange is found to be 0.295 V. The equilibrium constant of the reaction is approximately

(A) 9.51×10^8 (B) 10

(C) 1×10^{10} (D) 9.51×10^9

19. A silver cup is plated with silver by passing 965 coulombs of electricity, the amount of silver deposited is: (Mass no. of Ag = 108)

(A) 1.08 g (B) 1.002 g

(C) 108 g (D) 9.89 g

20. E° values of Mg^{2+}/Mg is –2.37 volts of Zn^{+2}/Zn is –0.76 volt and Fe^{+2}/Fe is –0.44 volt. Which of the following statements is correct?

(A) Zn oxidize Fe (B) Zn will reduce Fe^{2+}

(C) Zn will reduce Mg^{2+} (D) Mg oxidizes Fe

21. Lowering of vapour pressure ΔP_1 of aqueous solution of acid HA_1 and ΔP_2 of aqueous solution of acid HA_2 are related as $\Delta P_1 > \Delta P_2$. Correct relation is (Assume same concentration for both acids)

(A) $pK_{a_1} > pK_{a_2}$ (B) $pK_{a_1} < pK_{a_2}$

(C) $pK_{a_1} = pK_{a_2}$ (D) none of these

Space for rough work

22. The correct relationship between the boiling points of very dilute solutions of $AlCl_3(T_1)$, and $CaCl_2(T_2)$ having the same molar concentrations is
(A) $T_1 = T_2$ (B) $T_1 > T_2$
(C) $T_2 > T_1$ (D) $T_2 \geq T_1$

23. The vapour pressure of pure benzene at 25°C is 639.7 mm Hg and the vapour pressure of a solution of solute in benzene at 25°C is 631.9 mm Hg. The molality of the solution is
(A) 0.156 (B) 0.108
(C) 0.518 (D) 0.815

24. The stabilization of the dispersed phase in a lyophobic sol is due to
(A) the viscosity of the medium.
(B) the surface tension of the medium.
(C) affinity for the medium.
(D) the formation of an electrical layer between the two phase.

25. The process which is catalysed by one of the products is called
(A) acid-based catalyst. (B) auto catalyst.
(C) negative catalyst. (D) positive catalyst.

26. Which of the following statements is correct regarding the slag obtained during the extraction of a metal like copper or iron?
(A) The slag is lighter and has higher melting point than the metal.
(B) The slag is lighter and has lower melting point than the metal.
(C) The slag is heavier and has higher melting point than the metal.
(D) The slag is heavier and has lower melting point than the metal.

27. In oxoacids of chlorine, Cl–O bond contains
(A) $p\pi$–$p\pi$ bond (B) $d\pi$–$d\pi$ bond
(C) $p\pi$–$d\pi$ bond (D) $d\pi$–$f\pi$ bond

28. The compound that is not a Lewis acid is
(A) $BaCl_2$ (B) $AlCl_3$
(C) BCl_3 (D) $SnCl_4$

29. The formula of sodium nitroprusside is
(A) $Na_4[Fe(CN)_5NO_2]$ (B) $Na_2[Fe(CN)_5NOS]$
(C) $NaFe[Fe(CN)_6]$ (D) $Na_2[Fe(CN)_6NO_2]$

30. Amongst $[TiF_6]^{2-}$, $[CoF_6]^{3-}$, Cu_2Cl_2 & $[NiCl_4]^{2-}$. The colourless species are
(A) $[TiF_6]^{2-}$ & Cu_2Cl_2 (B) $[CoF_6]^{3-}$ & $[NiCl_4]^{2-}$
(C) $[TiF_6]^{2-}$ & $[CoF_6]^{3-}$ (D) Cu_2Cl_2 & $[NiCl_4]^{2-}$

Space for rough work

Mathematics PART – III

SECTION – A

Straight Objective Type

This section contains **30 multiple choice questions** numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. The number of values of a for which $\hat{\alpha} = a\hat{i} + 2a\hat{j} - 3a\hat{k}, \beta = (2a+1)\hat{i} + (2a+3)\hat{j} + (a+1)\hat{k}$, and $\hat{\gamma} = (3a+5)\hat{i} + (a+5)\hat{j} + (a+2)\hat{k}$ are coplanar
   (A) 0 (B) 1
   (C) 3 (D) 6

2. The differential equation $\left\{xy^3(1+\cos x) - y\right\}dx + xdy = 0$ represents the curve $\frac{x^2}{ay^2} = \frac{x^3}{b} + x^2\sin x + cx\cos x - d\sin x + k$. Then a + b + c + d is equal to
   (A) 8 (B) 9
   (C) 10 (D) 11

3. Let $\bar{p}$ and $\bar{q}$ be unit vectors such that $\bar{p}\times\bar{q} + \bar{p} = \bar{r}$ and $\bar{r}\times\bar{p} = \bar{q}$. Then the value of $[\bar{p}\ \ \bar{q}\ \ \bar{r}]$ is
   (A) 0 (B) 1
   (C) 2 (D) 3

4. The matrix $A = \begin{bmatrix} a & b & c \\ c & a & b \\ b & c & a \end{bmatrix}$ is an orthogonal matrix. Then $a+b+c = \pm k$, where k is
   (A) 0 (B) 1
   (C) 2 (D) 3

5. Seven digit numbers are formed using digits 1, 2, 3, 4, 5, 6, 7, 8, 9 without repetition. The probability of selecting a number such that product of any 5 consecutive digits is divisible by either 5 or 7 is $\frac{k}{12}$, then k is
   (A) 2 (B) 4
   (C) 6 (D) 7

Space for rough work

6. If a singular matrix $A = \left[a_{ij}\right]_{2\times 2}$ always commute with $B = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$ such that $\frac{a_{11}}{a_{12}} = \sqrt{k}$, then k is

(A) 0 (B) 1
(C) 2 (D) 3

7. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ be two matrices such that they are commutative and $c \neq 3b$ then the value of $\left|\frac{a-d}{3b-c}\right|$ is

(A) 0 (B) 1
(C) 2 (D) 3

8. Two natural numbers x and y are chosen at random. If P is the probability that $x^2 + y^2$ is divisible by 5 and 7 than the value of 25 P is

(A) 3 (B) 6
(C) 9 (D) none of these

9. $\left|\bar{p}\times\hat{i}\right|^2 + \left|\bar{p}\times\hat{j}\right|^2 + \left|\bar{p}\times\hat{k}\right|^2$ is $K\left|\bar{p}\right|^2$. Then K is

(A) 0 (B) 1
(C) 2 (D) 3

10. Given $\hat{a},\hat{b},\hat{c}$ are non – zero and coplanar then

(A) $\hat{a}-\hat{b},\hat{b}-\hat{c},\hat{c}-\hat{a}$ are always coplanar (B) $\hat{a}+\hat{b},\hat{b}+\hat{c},\hat{c}+\hat{a}$ are always coplanar
(C) $2\hat{a}+3\hat{b},2\hat{b}+3\hat{c},2\hat{c}+3\hat{a}$ are always coplanar (D) none of these

11. The lines $\frac{x-1}{2} = \frac{y}{-1} = \frac{z}{2}$ and $x - y + z - 2 = 0 = \lambda x + 3z + 5$ are coplanar for λ is equal to

(A) $-\frac{97}{11}$ (B) $-\frac{43}{5}$
(C) $-\frac{73}{9}$ (D) –5

12. If A is a square matrix of order n with |A| = 9 and |adj adj adj A| = 3^{16} then number of elements in A is

(A) 0 (B) 1
(C) 2 (D) 3

Space for rough work

13. For two unimodular complex numbers z_1 and z_2 $\begin{bmatrix} \bar{z}_1 & -z_2 \\ \bar{z}_2 & z_1 \end{bmatrix}^{-1}\begin{bmatrix} z_1 & z_2 \\ -\bar{z}_2 & \bar{z}_1 \end{bmatrix}^{-1} = k\begin{bmatrix} 1/4 & 0 \\ 0 & 1/4 \end{bmatrix}$ then k is

(A) 4 (B) 2
(C) 0 (D) 1

14. If a, b, c are the roots of $x^3 + 2x^2 + p = 0$, then the value of $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ is

(A) 2 (B) 4
(C) 6 (D) 8

15. A line L is passing through (1, 1, 1) intersects line $x = \frac{y}{2} = \frac{z}{3}$ at A and line $L_2 \equiv 2(x + 1) = 6(y + 2) = 3(z + 3)$ at B. Then distance of B from (0, 0, 0) is

(A) $\sqrt{12}$ (B) $\sqrt{14}$
(C) $\sqrt{18}$ (D) none of these

16. The statements are not true

(A) $\det(A+B) = \det A + \det B$ for all square matrices A, B of order n
(B) $\det(AB) = \det A.\det B$ for all A, B of order n
(C) $\det(A.B) = \det A \times \det B$
(D) none of these

17. There are n urns $u_1, u_2, \ldots, u_n$. Each urn contains $2n+1$ balls. The ith urn contains 2i number of balls $P(u_i)$, i.e., probability of selecting i^{th} urn is proportional to i^2+2. If we randomly select one of the urns and draw one ball and probability of ball being white be $P(A)$, then $\lim_{n\to\infty} P(A)$ is

(A) $\frac{3}{4}$ (B) $\frac{1}{4}$
(C) $\frac{2}{3}$ (D) $\frac{3}{16}$

Space for rough work

18. Let $\bar{p}, \bar{q}, \bar{r}$ be three vectors such that $[\bar{p}\ \ \bar{q}\ \ \bar{r}] = 2$. If $\bar{a} = x(\bar{q} \times \bar{r}) + y(\bar{r} \times \bar{p}) + z(\bar{p} \times \bar{q})$ is perpendicular to $\bar{p} + \bar{q} + \bar{r}$. Then the value of $x + y + z$ is
(A) 2 (B) 1
(C) 0 (D) none of these

19. PQRSP'Q'R'S' (where P′, Q′, R′, S′ are vertices corresponding to P, Q, R, S respectively) is a cube of edges 1 unit. A and B are midpoints of the edges Q'P' and Q'R' respectively. The distance of vertex S from the plane AQB is
(A) $\frac{3}{4}$ units (B) $\frac{3}{2}$ units
(C) $\frac{4}{3}$ units (D) 1 unit

20. If $\vec{a}, \vec{b}, \vec{c}$ are unit vectors perpendicular to each other such that $\vec{r}.\vec{a} = 1$, $\vec{r}.\vec{b} = 2$ and $\vec{r}.\vec{c} = 3$ then
(A) $\vec{r} = 3\vec{a} + 2\vec{b} + \vec{c}$ (B) $\vec{r} = \vec{a} + 2\vec{b} + 3\vec{c}$
(C) $\vec{r} = 3\vec{a} \times \vec{b} + 2\vec{b} \times \vec{c} + \vec{c} \times \vec{a}$ (D) none of these

21. Four of the eight vertices of a regular octagon are chosen at random. The probability that the figure with these four vertices is square, equals
(A) $\frac{1}{14}$ (B) $\frac{1}{7}$
(C) $\frac{1}{28}$ (D) $\frac{1}{35}$

22. The differential equation $\frac{dx}{dy} = \frac{\sqrt{1-x^2}}{x}$ determines a family of circles with
(A) variable radii and a fixed centre (1, 0)
(B) variable radii and a fixed centre (–1, 0)
(C) fixed radius 1 and variable centres along x-axis
(D) fixed radius 1 and variable centres along y-axis

23. If the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-4}{1} = \frac{y-6}{2} = \frac{z+k}{2}$ intersect then the value of k is
(A) – 7 (B) – 8
(C) – 9 (D) – 10

Space for rough work

24. The value of determinant $\Delta = \begin{vmatrix} 1 & \frac{4\sin B}{b} & \cos A \\ 2a & 8\sin A & 1 \\ 3a & 12\sin A & \cos B \end{vmatrix}$ is (where a, b and c are the sides of a triangle and A, B and C are angles opposite to a, b and c respectively)

(A) 0 (B) 1

(C) 2 (D) 3

25. The equations $ax + a^2y + z = 0$, $bx + b^2y + z = 0$, $cx + c^2y + z = 0$ have only zero solution if common ratio of G.P. is not equal to (here a, b, c are in G.P.)

(A) $0, \pm 1$ (B) $2, \pm 3$

(C) $4, \pm 5$ (D) none of these

26. The value of $\Delta = \begin{vmatrix} \lim\limits_{x\to 0} \tan\left(\frac{\pi}{4} + x\right)^{1/x} & x^3 & e^2 \\ \lim\limits_{x\to\infty} (1)^x & x^4 + 1 & 1 \\ \lim\limits_{x\to 0} \frac{x^2 \sin\frac{1}{x}}{\sin x} & x^5 + 9 & 0 \end{vmatrix}$ equals

(A) 1 (B) 2

(C) 3 (D) none of these

27. If $\hat{a}$, $\hat{b}$, $\hat{c}$ are three non-coplanar unit vectors, then $\left[\hat{a} \quad \vec{p} \quad \vec{q}\right] \hat{a} + \left[\hat{b} \quad \vec{p} \quad \vec{q}\right] \hat{b} + \left[\hat{c} \quad \vec{p} \quad \vec{q}\right] \hat{c}$ is equal to

(A) $\left(\hat{a} + \hat{b} + \hat{c}\right) \times \left(\vec{p} \times \vec{q}\right)$ (B) $\hat{a} + \hat{b} + \hat{c} + \vec{p} + \vec{q}$

(C) $\vec{p} + \vec{q}$ (D) $\vec{p} \times \vec{q}$

Space for rough work

28. If $\vec{a}$ and $\vec{b}$ are two unit vectors and θ is the angle between them, then the unit vector along the angular bisector of $\vec{a}$ and $\vec{b}$ will be given by

(A) $\dfrac{\vec{a}-\vec{b}}{2\cos\dfrac{\theta}{2}}$ (B) $\dfrac{\vec{a}+\vec{b}}{2\cos\dfrac{\theta}{2}}$

(C) $\dfrac{\vec{a}+\vec{b}}{2\cos\dfrac{\theta}{2}}$ (D) none of these

29. The equation of the line passing through the point (1, –2, 3) and parallel to the line $x - y + 2z = 5$ and $3x + y + z = 6$ is

(A) $\dfrac{x-1}{-3}=\dfrac{y+2}{5}=\dfrac{z-3}{4}$ (B) $\dfrac{x-1}{1}=\dfrac{y+2}{-1}=\dfrac{z-3}{2}$

(C) $\dfrac{x-1}{3}=\dfrac{y+2}{1}=\dfrac{z-3}{6}$ (D) $\dfrac{x-1}{3}=\dfrac{y+2}{-1}=\dfrac{z-3}{2}$

30. The area of the region bounded by $x=\dfrac{1}{2}$, $x = 2$, $y = \ln x$ and $y = 2^x$ is

(A) $\dfrac{3}{2}-\dfrac{5}{2}\ln 2+\dfrac{4-\sqrt{2}}{\ln 2}$ (B) $\dfrac{3}{2}+\dfrac{5}{2}\ln 2+\dfrac{4-\sqrt{2}}{\ln 2}$

(C) $\dfrac{3}{2}+\dfrac{5}{2}\ln 2+\dfrac{4+\sqrt{2}}{\ln 2}$ (D) $\dfrac{3}{2}-\dfrac{5}{2}\ln 2-\dfrac{4-\sqrt{2}}{\ln 2}$

Space for rough work